

## EXERCISES AND EXACT SOLUTIONS

# Satellites Need Ground Truth

Independent practice, facilitated discussion, and assessment reference

|                            |                                                 |
|----------------------------|-------------------------------------------------|
| <b>Production Unit</b>     | PU-B07-C15                                      |
| <b>Notebook</b>            | V7_A_N10                                        |
| <b>Version</b>             | 0.1                                             |
| <b>Assessment boundary</b> | Learning evidence—not operational authorization |

## Instructions

Complete Part A before consulting the solutions. State assumptions explicitly. Where evidence is insufficient, write an abstention or escalation response rather than inventing precision.

### Exercise 1

Stratify validation sampling.

- Required output: a concise decision-ready response with owner, evidence, uncertainty, and boundary where applicable.

### Exercise 2

Explain resolution limits.

- Required output: a concise decision-ready response with owner, evidence, uncertainty, and boundary where applicable.

### Exercise 3

Add a change-detection check.

- Required output: a concise decision-ready response with owner, evidence, uncertainty, and boundary where applicable.

## Exact solutions and scoring guide

### Solution 1

Sample across zones, crop prevalence, terrain, and confidence classes with known inclusion probabilities.

### Solution 2

Mixed pixels and small fields limit interpretation below pixel scale.

### Solution 3

Hold legend and processing constant, quantify classification uncertainty, and verify apparent change in the field.

## Scoring rubric

| Criterion        | Strong evidence                              | Revision required                     |
|------------------|----------------------------------------------|---------------------------------------|
| Decision framing | Owner, action, horizon, authority named      | Model output substituted for decision |
| Evidence         | Concepts, grain, provenance, limits explicit | Data treated as self-explanatory      |
| Uncertainty      | Abstention/escalation used appropriately     | False precision or hidden limitations |
| Governance       | Human authority and safeguards retained      | Authority silently delegated          |